

Constrained Optimization



Prithwijit Guha
Dept. of EEE, IIT Guwahati

Overview

- The Constrained Optimization Problem
- Contour Plot and Gradients
- Objective Function and Gradients
- The KKT Conditions
 - Necessary Conditions
 - Sufficient Conditions
- The Primal-Dual Formulation

Constrained Optimization: Definition & Formulation

Constrained Optimization is the Process of Maximizing or Minimizing a Desired Objective Function while Satisfying a Set of Prevailing Constraints

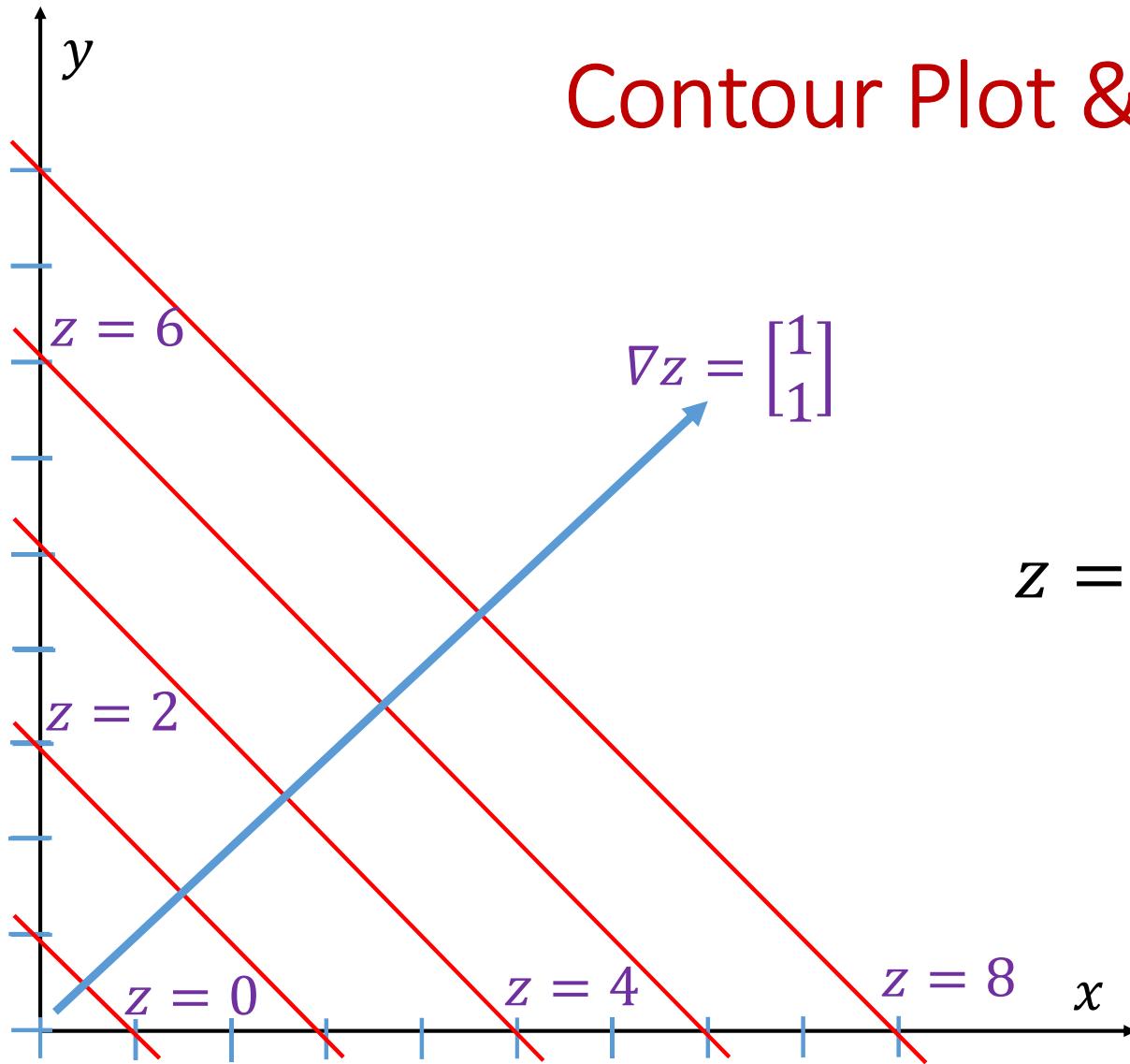
Minimize $f(X)$

Subject To

$$g_i(X) \leq 0; i = 1, \dots, m$$

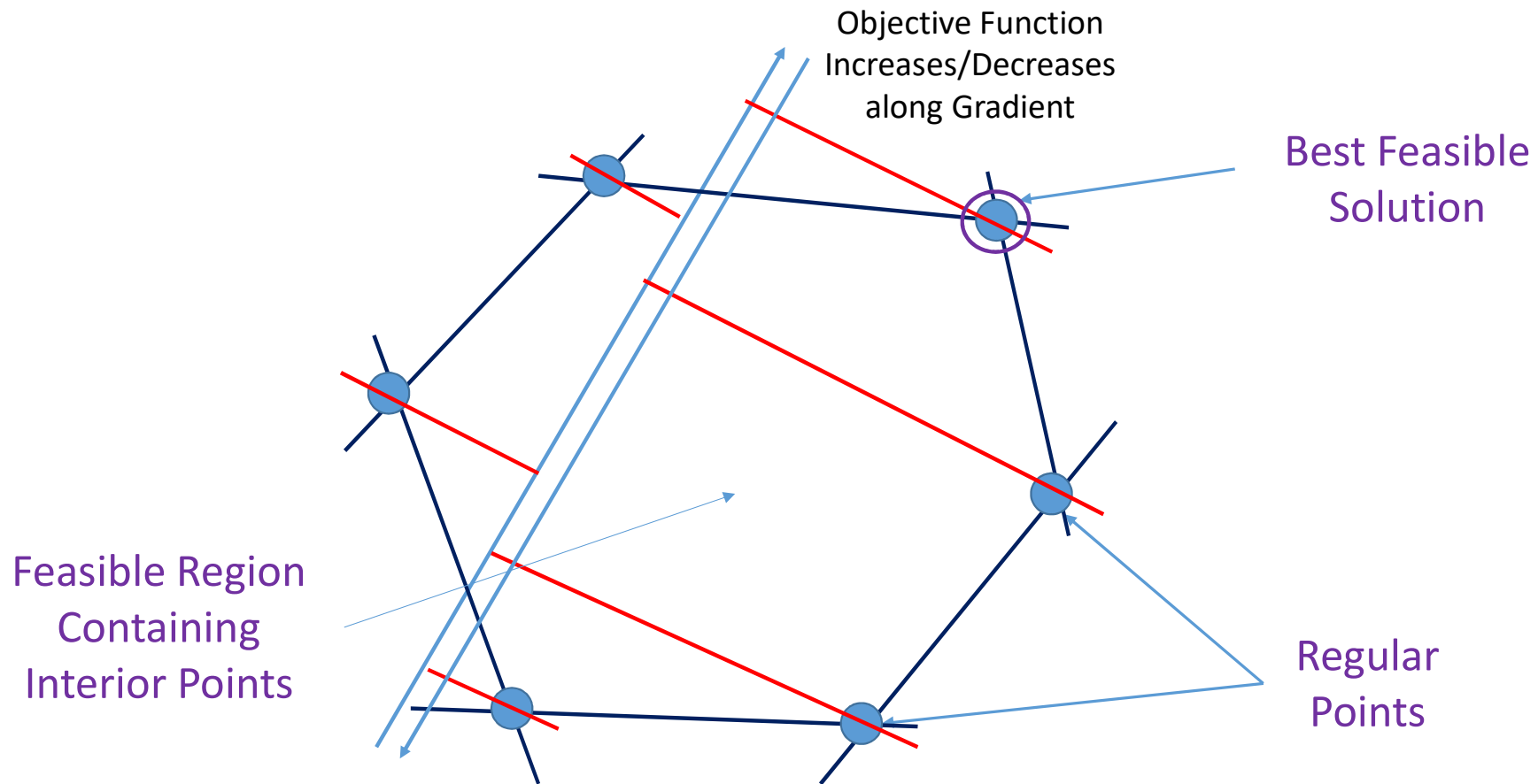
$$h_j(X) = 0; j = 1, \dots, l$$

Contour Plot & Gradient

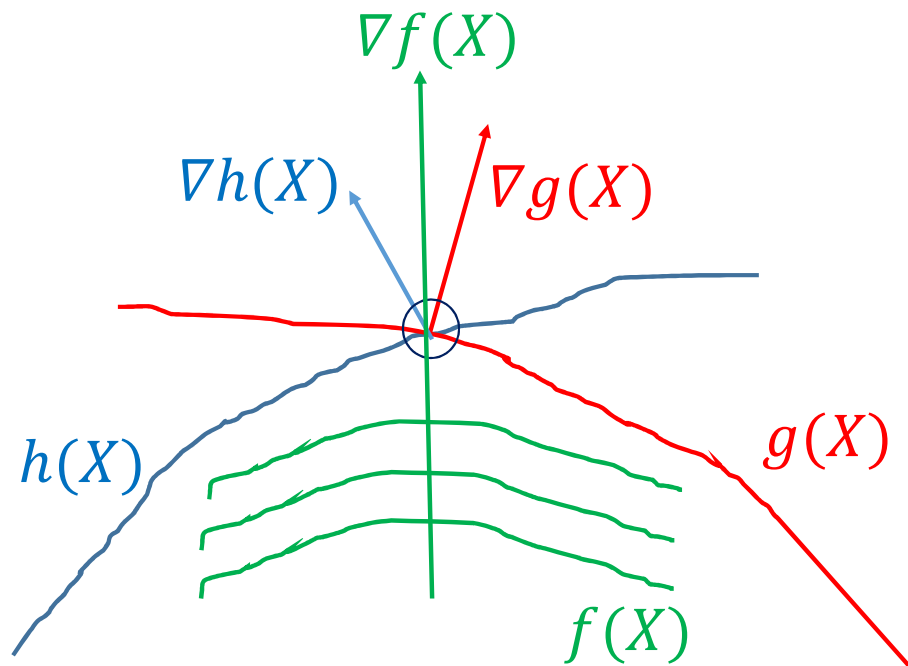


$$z = f(x, y) = x + y - 1$$

Objective Functions & Gradients: Linear Case



Objective Functions & Gradients (Contd.)



Minimize $f(X)$

Subject to $g(X) \leq 0$

$h(X) = 0$

Observation:

*$\nabla f(X)$, $\nabla g(X)$ and $\nabla h(X)$
pass through the same point*

Thus, we can write

$$\nabla f(X) + \mu \nabla g(X) + \lambda \nabla h(X) = 0$$

Karush-Kuhn-Tucker (KKT) Conditions

Minimize $f(X)$ subject to $g_i(X) \leq 0; i = 1 \dots m$ and $h_j(X) = 0; j = 1 \dots l$

Construct Lagrangian Function L

$$L(X; \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(X) + \sum_{i=1}^m \mu_i g_i(X) + \sum_{j=1}^l \lambda_j h_j(X)$$

Where, $\boldsymbol{\mu} = \{ \mu_1 \dots \mu_i \dots \mu_m \}$ and $\boldsymbol{\lambda} = \{ \lambda_1 \dots \lambda_j \dots \lambda_l \}$

KKT Conditions (Contd.)

Optimality: $\nabla_X L(X; \mu, \lambda) = \nabla_X f(X) + \sum_{i=1}^m \mu_i \nabla_X g_i(X) + \sum_{j=1}^l \lambda_j \nabla_X h_j(X) = 0$

Non – negativity: $\mu_i \geq 0 ; i = 1 \dots m$

Complementarity: $\mu_i g_i(X) = 0 ; i = 1, \dots m$

Feasibility: $g_i(X) \leq 0 ; i = 1, \dots m$ and $h_j(X) = 0 ; j = 1, \dots l$

Necessary Conditions

KKT Conditions (Contd.)

X^* : Optimal Solution of $\nabla_X L(X; \mu, \lambda) = 0$

Hessian Matrix $\nabla_X^2 L(X^*; \mu, \lambda)$ is Positive Definite in $\mathbf{T} \subset \mathbb{R}^n$

$$\mathbf{T} = \{y: [\nabla g_i(X^*)^T y = 0; i = 1 \dots m] \wedge [\nabla h_j(X^*)^T y = 0; j = 1 \dots l]\}$$

Sufficient Condition: $y^T \nabla_X^2 L(X^*; \mu, \lambda) y > 0; \|y\|_2 \neq 0; y \in \mathbf{T}$

Stronger Condition: $y^T \nabla_X^2 L(X^*; \mu, \lambda) y > 0; \|y\|_2 \neq 0; y \in \mathbb{R}^n$

Problem

$$\text{Minimize } f(x_1, x_2) = 2x_1 + x_2$$

Subject To

$$h(x_1, x_2) = x_1^2 + x_2^2 - 1 = 0$$

Solution

$$L(x_1, x_2; \lambda) = 2x_1 + x_2 + \lambda(x_1^2 + x_2^2 - 1)$$

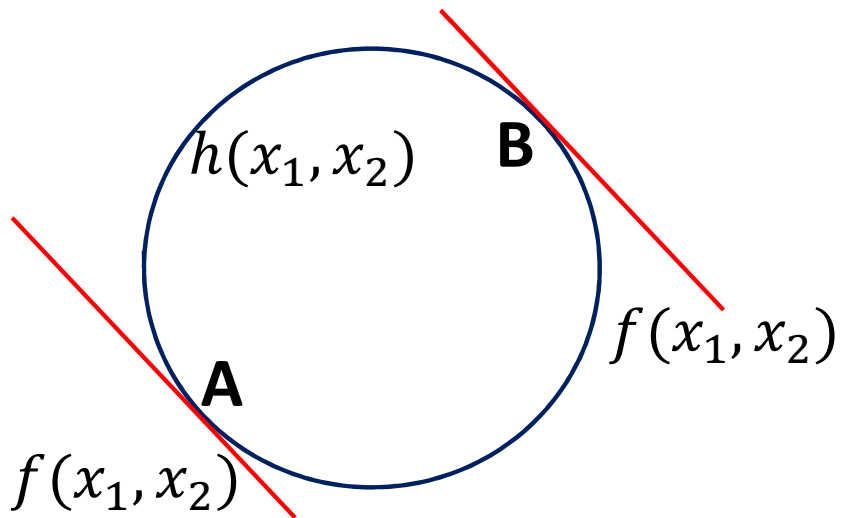
$$\frac{\partial L}{\partial x_1} = 0 \rightarrow x_1 = -\frac{1}{\lambda}$$

$$\frac{\partial L}{\partial x_2} = 0 \rightarrow x_2 = -\frac{1}{2\lambda}$$

$$\lambda = \pm \frac{\sqrt{5}}{2}$$

$$\rightarrow A = \left(x_1 = -\frac{2}{\sqrt{5}}, x_2 = -\frac{1}{\sqrt{5}} \right)$$

$$\rightarrow B = \left(x_1 = \frac{2}{\sqrt{5}}, x_2 = \frac{1}{\sqrt{5}} \right)$$



Primal-Dual Formulation

Minimize $f(X)$ subject to $g_i(X) \leq 0; i = 1 \dots m$ and $h_j(X) = 0; j = 1 \dots l$

X^*, μ^*, λ^* : Optimal Solution of the **Primal Problem** $\nabla_X L(X; \mu, \lambda) = 0$

(X^*, μ^*, λ^*) is a **Saddle Point** of $L(X; \mu, \lambda)$ i.e.

$$L(X^*, \mu, \lambda) \leq L(X^*, \mu^*, \lambda^*) \leq L(X, \mu^*, \lambda^*)$$

(X^*, μ^*, λ^*) solves the following dual problem

$$\text{Maximize } L = f + \mu^T g + \lambda^T h$$

$$\text{Subject To } [\nabla L = 0] \wedge [\mu \geq 0]$$

Problem

$$\text{Minimize } f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 3)^2$$

Subject To

$$g_1(x_1, x_2) = 2x_1 + x_2 - 2 \leq 0$$

$$g_2(x_1, x_2) = -x_1 \leq 0$$

$$g_3(x_1, x_2) = -x_2 \leq 0$$

Solution

$$L(x_1, x_2; \boldsymbol{\mu}) = (x_1 - 3)^2 + (x_2 - 3)^2 + \mu_1(2x_1 + x_2 - 2) - \mu_2 x_1 - \mu_3 x_2$$

$$\frac{\partial L}{\partial x_1} = 0 \rightarrow x_1 = \frac{\mu_2}{2} - \mu_1 + 3$$

$$\frac{\partial L}{\partial x_2} = 0 \rightarrow x_2 = \frac{\mu_3}{2} - \frac{\mu_1}{2} + 3$$

$$Q(\boldsymbol{\mu}) = -\frac{5\mu_1^2}{4} - \frac{\mu_2^2}{4} - \frac{\mu_3^2}{4} + \frac{\mu_1\mu_3}{2} + \mu_1\mu_2 + 7\mu_1 - 3\mu_2 - 3\mu_3$$

Dual Function to be Maximized w. r. t. $\boldsymbol{\mu}$

Applications

- Principal Component Analysis
- Linear Discriminant Analysis
- Factor Analysis
- Regularization
- Neural Network Learning
- Support Vector Machines
- Parameter Estimation And Many Others...

Summary

- Constrained Minimization
- Visualizations
 - Contour Plots & Gradients
 - Objective Functions & Gradients
 - Feasible Regions and Best Solutions
- Karush-Kuhn-Tucker Conditions
 - Necessary Conditions
 - Sufficient Conditions
- Primal-Dual Formulations
- Applications



Thank You